ENTERPRISE RESOURCES PLANNING SOFTWARE WITH POS

Submitted by

Abu Sufian Emon	Omor Faruque Tanim	Mashiath Chowdhury
ID: 0562210005101023	ID: 0562210005101004	ID: 0562210005101030

Supervised by

Muthmainna Mou

Lecturer

Department of Computer Science and Engineering

B.Sc. (Engg.)

IN

COMPUTER SCIENCE AND ENGINEERING



Department of Computer Science and Engineering

North East University Bangladesh (NEUB)

Sylhet, Bangladesh

February, 2026

ENTERPRISE RESOURCES PLANNING SOFTWARE WITH POS



A work submitted to the Department of Computer Science and Engineering, North East University Bangladesh, for partial fulfillment of the requirements for the degree of B.Sc. (Engg.)in Computer Science and Engineering

Submitted by

Abu Sufian Emon	Omor Faruque Tanim	Mashiath Chowdhury
ID: 0562210005101023	ID: 0562210005101004	ID: 0562210005101030

Supervised by

Muthmainna Mou
Lecturer
Department of Computer Science and Engineering

February, 2026

Qualification Form of B.Sc. (Engg.) degree

This project titled, **"Enterprise Resources Planning Software with POS"**, submitted by Abu Sufian Emon (ID: 0562210005101023), Omor Faruque Tanim (ID: 0562210005101004), Mashiath Chowdhury (ID: 0562210005101030), Session: February 2026, has been accepted as satisfactory in partial fulfillment of the requirement for the degree of B.Sc. (Engg.) in Computer Science and Engineering on February, 2026.

BOARD OF EXAMINERS

Dr. Arif Ahmad
Associate Professor Head
Department of Computer Science and Engineering

Muthmainna Mou
Lecturer
Department of Computer Science and Engineering

Candidate's Declaration

This is to certify that the work presented in this project entitled, "Enterprise Resources Planning Software with POS", is the outcome of the research carried out by Abu Sufian Emon (ID: 0562210005101023), Omor Faruque Tanim (ID: 0562210005101004), Mashiath Chowdhury (ID: 0562210005101030), under the supervision of Muthmainna Mou, Lecturer, Department of Computer Science and Engineering, North East University Bangladesh (NEUB), Sylhet, Bangladesh.

It is also declared that neither this thesis nor any part thereof has been submitted anywhere else for the award of any degree, diploma, or other qualifications.

Signature of the Candidate

Abu Sufian Emon
ID: 0562210005101023

Omor Faruque Tanim
ID: 0562210005101004

Mashiath Chowdhury
ID: 0562210005101030

Dedication

We dedicate this project to our families, friends, and mentors, whose support, encouragement, and guidance have been invaluable throughout the development of this work.

Contents

Certification iii

Candidate's Declaration iv

Dedication v

List of Figures viii

List of Tables xi

List of Abbreviations xii

Acknowledgement xiii

Abstract xiv

1 Introduction 1

1.1 Project Overview 1
1.2 Problem Statement 1
1.3 Objectives 2
1.4 Project Motivation 2
1.5 Target User Groups 2
1.6 Similar Products and Their Features 3
1.6.1 Odoo 3
1.6.2 SAP Business One 3
1.6.3 Microsoft Dynamics 365 3
1.7 Features of Our Product as Requirements 4
1.7.1 Functional Requirements 4
1.7.2 Non-Functional Requirements 4

2 Planning 6

2.1 User Roles and Descriptions 6
2.2 User Stories 7

2.3 Scrum Increment 8
2.3.1 Sprint Work Distribution 9

3 Methodology **10**
3.1 System Architecture 10
3.2 Database Design 10
3.3 Operational Workflow 13

4 Results and Discussion **15**
4.1 Functional Achievement and Evaluation 15
4.1.1 Operational Efficiency 15
4.1.2 Security and Access Control 15
4.2 Project Screenshots and Explanation 16
4.2.1 CEO Interface 17
4.2.2 Branch Manager Interface 20
4.2.3 Executive User Interface 24
4.2.4 Inventory Manager Interface 30
4.2.5 Accounts Interface 36

5 Conclusion **40**
5.1 Project Summary 40
5.2 Project Contributions and Practical Impact 40
5.3 Final Closing Statement 41

List of Figures

3.1 Detailed ER Diagram showing the relationships between Inventory, Sales, and Branch entities. 12
3.2 Use Case Diagrams 14

4.1 User Registration Interface and Information Submission Confirmation . 16
4.2 CEO Approval Interface 17
4.3 CEO Panel: Financial Summaries, Branch Status, and Alert Management 17
4.4 Active User List with Role and Branch Designations 17
4.5 Centralized Branch List with Management Controls 18
4.6 Multi-branch Revenue Report 18
4.7 Detailed Sales Report and Transaction Logs 18
4.8 Consolidated Expense Report for Organizational Cost Analysis 19
4.9 Profit and Loss Report for Organizational Performance Evaluation . . . 19
4.10 CEO Interface for Reviewing and Authorizing Inter-Branch Stock Transfers 19
4.11 Monitoring View for Stock Transfers Awaiting Final Receipt Confirmation 20
4.12 Current Inventory Levels for the Specific Branch 20
4.13 Consolidated View of Stock Levels Across All Branches 20
4.14 Interface for Adding New Customer Records 21
4.15 List of Registered Customers and Contact Information 21
4.16 Management Interface for Authorizing Retail Sales 21
4.17 Generated Invoice Template for Completed Sales 22
4.18 Interface for Creating Corporate Sales Quotations 22
4.19 Internal Request System for Procurement of New Items 22
4.20 Tracking Interface for Incoming Inter-Branch Transfers 23
4.21 Overview of Pending and Fulfilled Product Requests 23
4.22 Detailed History of Stock Movements Between Branches 23
4.23 Submission Form for Local Branch Expenditures 24
4.24 Tracking and Categorization of Branch-Level Expenses 24
4.25 Point-of-Sale interface for retail transactions 24
4.26 Branch-specific retail transaction history 25

4.27 Standard invoice generation for customers 25
4.28 Interface for updating transaction payment status 26
4.29 Management list for registered retail customers 26
4.30 Verified list of corporate account entities 26
4.31 Generation form for corporate price quotations 27
4.32 Branch record of all issued corporate quotes 27
4.33 Interface for processing corporate-tier sales 27
4.34 Log of billing cycles for corporate clients 28
4.35 Tracking of outstanding corporate invoices 28
4.36 Current inventory availability for the local branch 28
4.37 Stock levels across all organizational branches 29
4.38 Detailed analytics of retail sales performance 29
4.39 Performance report for corporate sales accounts 29
4.40 Summary of cash flow and daily collections 30
4.41 Dashboard featuring critical stock alerts and inventory status 30
4.42 Master list of all products registered in the system 30
4.43 Form for adding new items to the inventory database 31
4.44 Interface for creating and organizing product categories 31
4.45 Overview of all active product classification categories 31
4.46 Record of all invoices received from external suppliers 32
4.47 Detailed view of specific supplier billing documents 32
4.48 Tool for updating paid amounts on supplier invoices 32
4.49 Interface for adding new supplier contact and billing data 33
4.50 Database of all approved organizational suppliers 33
4.51 System for generating new stock purchase orders 33
4.52 Tracking for orders awaiting warehouse arrival 34
4.53 Interface for receiving stock or processing returns to suppliers 34
4.54 Logged record of all items returned to suppliers 34
4.55 Real-time monitoring of global stock movements 35
4.56 Logs of damaged or defective inventory items 35
4.57 Entry form for documenting stock discrepancies or damage 35
4.58 Interface for initiating stock transfers between branches 36
4.59 Management of incoming stock requests from various branches 36
4.60 Financial overview featuring key liquidity metrics and alerts 36
4.61 Review queue for authorizing new corporate account heads 37
4.62 List of customers awaiting account verification or credit approval 37
4.63 Interface for auditing and editing customer financial statements 37
4.64 Real-time monitoring of physical cash available at each branch 38

4.65 Tracking of bank deposits and cash transfers from branches 38
4.66 Consolidated audit of all organizational expenditures 38
4.67 Tracking of pending payments from all corporate clients 39
4.68 Global report of total sales versus outstanding receivables 39

List of Tables

2.1 User Roles and Responsibilities 6
2.2 Parallel development tasks by user role and sprint. 9

List of Abbreviations

CEO	Chief Executive Officer
CSV	Comma-Separated Values
ERP	Enterprise Resource Planning
Executive	Executive Officer
Manager	Project/Team Manager
POS	Point of Sale
SQL	Structured Query Language

Acknowledgement

We, the project team, would like to express our sincere gratitude to all those who have contributed to the successful completion of our ERP System project.

First and foremost, we extend our heartfelt thanks to our supervisor, Muthmainna Mou for their invaluable guidance, constructive feedback, and continuous support throughout the development process.

We are deeply grateful to our department and faculty members for providing us with the resources, knowledge, and encouragement that have enabled us to carry out this work.

Our families and friends deserve special appreciation for their constant motivation, patience, and understanding during the long hours of research, coding, and documentation.

Finally, we acknowledge the collaborative spirit of our team members, whose dedication, hard work, and commitment have been the driving force behind the success of this project.

Abstract

Abstract

Enterprise Resource Planning (ERP) systems have become essential tools for organizations aiming to integrate complex business processes, improve operational efficiency, and support data-driven decision-making. This project focuses on the design and development of a centralized, web-based ERP solution tailored for multi-branch retail and corporate environments. The system streamlines core operations by providing role-specific modules for the **CEO, Inventory Manager, Branch Manager, Accounts, and Executive**.

The proposed system is built using a relational database architecture to manage the full lifecycle of inventory and sales. Key functionalities include multi-branch stock tracking, inter-branch product transfers with approval workflows, management of faulty items, and a dual-stream sales pipeline for both retail and corporate clients. By integrating these processes into a single platform, the system ensures data consistency across geographically distributed branches, such as Sylhet and Rangpur.

To evaluate the system, we conducted testing using real-world inventory datasets and multi-role operational scenarios. The results indicate significant improvements in stock visibility, a reduction in manual reconciliation errors between branches, and enhanced financial traceability regarding supplier dues and branch-wise expenses. This ERP solution provides a robust foundation for SMEs to automate fragmented processes, and it is designed for future scalability, including the integration of advanced data visualization and mobile accessibility to meet evolving business needs.

Keywords: ERP, Inventory Management, Multi-branch Operations, Business Process Automation, Relational Database.

Chapter 1

Introduction

1.1 Project Overview

Enterprise Resource Planning (ERP) systems are critical tools for modern businesses, enabling them to manage operations, resources, and communications efficiently in a unified platform. Our ERP System is a web-based application designed to centralize and automate company processes for organizations with multiple roles and hierarchical structures. It supports executives, branch managers, inventory specialists, and the CEO in carrying out their daily responsibilities effectively.

The system provides a seamless workflow from multi-channel product sales (Direct and Corporate) to invoice management, inventory tracking, branch operations, and company-wide reporting. Its role-based access ensures that each user can perform tasks specific to their responsibilities, minimizing complexity and improving operational efficiency across geographically distributed branches.

1.2 Problem Statement

Many organizations still rely on manual or semi-digital processes for sales recording, stock management, and branch-level expense tracking. In a multi-branch environment, this leads to:

- Time-consuming manual data entry and delayed report preparation.
- Inability to track real-time stock levels, leading to overstocking or stock-outs.

- Lack of transparency in inter-branch product transfers and ”faulty item” accounting.
- Communication delays between regional branches (e.g., Sylhet, Rangpur) and headquarters.
- Difficulty in managing complex corporate quotation and credit collection cycles.

1.3 Objectives

The primary objectives of this ERP system are:

1. To create a centralized database supporting multiple user roles (Executive, Branch Manager, Inventory Manager, and CEO).
2. To automate professional invoice generation and payment tracking for both retail and corporate clients.
3. To implement a robust inventory lifecycle, including procurement, inter-branch transfers, and faulty product management.
4. To provide real-time financial oversight through branch-wise profit/loss and expense monitoring.

1.4 Project Motivation

In many organizations, operations such as sales tracking and branch-level expense monitoring are handled through fragmented software systems. This often leads to data inconsistency and difficulty in generating consolidated reports for stakeholders. Our motivation for building this ERP system was to address these challenges by creating a single, integrated platform that ensures accurate, real-time data flow from the point of sale to the final financial audit.

1.5 Target User Groups

- **Executives** – Manage daily sales, register customers, generate invoices, and track personal performance.

- **Branch Managers** – Oversee local branch operations, approve local transactions, and manage branch-specific expenses.
- **Inventory Managers** – Handle procurement from suppliers, manage lot numbers, oversee branch transfers, and track faulty stock.
- **Accounts Personnel** – Reconcile supplier dues, monitor cash deposits, and verify operational expenditures.
- **CEO** – Access high-level strategic dashboards, approve new system users, and monitor company-wide revenue.

1.6 Similar Products and Their Features

Several ERP solutions exist in the market, such as Odoo, SAP Business One, and Microsoft Dynamics 365. While these products are powerful, they often lack tailored workflows for localized SMEs or come at costs that are prohibitive.

1.6.1 Odoo

Odoo offers modular ERP features including sales and inventory, but its open-source version requires significant technical expertise to customize for specific inter-branch transfer workflows.

1.6.2 SAP Business One

This platform provides comprehensive business management features but comes with a steep learning curve and high licensing costs that are often not feasible for growing businesses in developing regions.

1.6.3 Microsoft Dynamics 365

A cloud-based ERP with advanced analytics, but it may be overly complex for organizations that require a lightweight, fast-loading interface for daily retail operations.

1.7 Features of Our Product as Requirements

1.7.1 Functional Requirements

- **Sales Management:** Facilitates a complete transaction lifecycle, supporting immediate Direct Sales for retail customers and a structured Corporate pipeline that transitions from initial Quotation to final Invoice and payment collection.
- **Branch Operations:** Empowers Branch Managers with localized autonomy to monitor real-time stock availability, initiate restock requests to the Inventory Manager, and maintain a digital ledger of all branch-specific operational expenses.
- **CEO Oversight:** Responsible for controlling user access through registration approvals, monitoring branch conditions and performance, and maintaining high-level supervision over the Accounts and Inventory departments.
- **Inventory Control:**
  - Add products and generate supplier invoices.
  - Manage outstanding supplier dues and lot-based tracking.
  - Facilitate and approve inter-branch stock transfers.
  - Track faulty/damaged items and calculate associated losses.
- **Financial Accounting:**
  - Monitor organization-wide cash flow and branch-specific cash-in-hand.
  - Manage and customer ledgers and corporate billing cycles.
  - Approve corporate heads and verify regional cash deposits.
  - Generate consolidated master reports for sales, dues, and expenditures.

1.7.2 Non-Functional Requirements

- **Usability:** Intuitive, role-based navigation requiring minimal training for branch staff.
- **Performance:** Optimized database queries for fast retrieval of sales and inventory data.

- **Security:** Strict Role-Based Access Control (RBAC) to prevent unauthorized financial access.
- **Scalability:** Designed to support an increasing number of branches and product categories.

Chapter 2

Planning

Effective planning is essential for the successful implementation of the Enterprise Resource Planning (ERP) system. This chapter defines the specific user roles and their responsibilities, presents the user stories tailored to the inventory and sales lifecycle, and outlines the Agile Scrum approach used for the project, including the concept of Scrum Increment.

2.1 User Roles and Descriptions

The ERP system involves five distinct user roles, structured to manage a multi-branch business environment. The table below summarizes these roles and their key responsibilities based on the system architecture.

Table 2.1: User Roles and Responsibilities

Role	**Key Responsibilities**
CEO	High-level oversight, user approval management, and accessing company-wide financial reports.
Inventory Manager	Product procurement, supplier management, lot-based stock tracking, and inter-branch transfers.
Branch Manager	Monitoring local branch stock, recording daily expenses, and approving local operational tasks.
Accounts	Managing supplier invoices (Dues), reconciling cash deposits, and auditing branch expenditures.
Executive	Point-of-Sale (POS) operations, customer registration, and retail invoice generation.

2.2 User Stories

The system requirements are modeled using a user-centered design approach. These stories define the functional boundaries for the five primary stakeholders, ensuring that every feature maps directly to a business need.

CEO (Strategic Oversight)

"As a CEO, I want to control system access and monitor the organizational health across all branches so that I can ensure security and profitability."

- **Acceptance Criteria:**
  - Ability to approve or decline pending user registration requests (`u_request.php`).
  - Real-time access to consolidated Revenue, Profit, and Loss reports.
  - Visibility into the "Pending Approval" and "Pending Receive" status of inter-branch stock movements.

Inventory Manager (Supply Chain Control)

"As an Inventory Manager, I want to manage the global product lifecycle and coordinate logistics so that I can prevent inventory imbalances across the distribution network."

- **Acceptance Criteria:**
  - CRUD (Create, Read, Update, Delete) functionality for products, categories, and suppliers.
  - Initiation of stock transfers from HQ to regional branches (e.g., Rangpur, Sylhet).
  - Capability to update supplier invoice due amounts and generate purchase documentation.

Branch Manager (Operational Supervision)

"As a Branch Manager, I want to monitor my local stock and record branch-specific expenditures so that my branch's net performance is accurately reported."

- **Acceptance Criteria:**
  - Dashboard restricted to displaying stock levels for the assigned branch only.
  - Interface to record operational expenses such as electricity, rent, and local courier charges.
  - System triggers to receive and verify incoming stock transfers.

  - Requests for product to other branches.
  - View Transfer history.
  - Customer and corporate head creation.

Accounts (Financial Audit)

"As an Accountant, I want to track company-wide payables and cash flow so that the organization remains financially stable."

- **Acceptance Criteria:**
  - Access to detailed expense reports filtered by branch and date.
  - Credit limit control and management.
  - Approve corporate heads.
  - Monitoring cash-in-hand branch-wise.
  - Monitor corporate bill collection pendings.
  - Monitor customer due bills.
  - Verification of cash deposits and revenue streams generated by the Executive department.

Executive (Front-line Operations)

"As an Executive, I want an efficient Point of Sale (POS) interface so that I can process customer transactions and maintain accurate sales records."

- **Acceptance Criteria:**
  - Automated calculation of totals and taxes during the sale process.
  - Immediate generation of downloadable and printable retail invoices.
  - Bulk order corporate quotation and management.
  - Processing and tracking of Corporate sales.
  - Corporate bill collection and payment logging.
  - Monitoring of corporate bill collection pending status.
  - Real-time tracking of customer due bills.
  - Generation of branch-specific cash-in-hand reports.
  - Maintenance of bank deposit records.

2.3 Scrum Increment

The development of this ERP system follows the **Agile Scrum** methodology. Work is divided into iterative **Sprints**, where each developer focuses on specific user-centric

modules to ensure parallel progress and specialized focus across the organization's functional needs.

2.3.1 Sprint Work Distribution

The following table outlines the parallel development tracks handled by the three-member development team over four sprints:

Sprint	**Omor Faruque Tanim (Accounts & Executive)**	**Mashiath Chowdhury (CEO & Branch Manager)**	**Abu Sufian Emon (Inventory Manager)**
Sprint 1	Requirement analysis for Corporate Billing & POS.	CEO Dashboard wireframing and User Role architecture.	Product Database schema and Category management.
Sprint 2	Development of **Direct Sales POS** and Customer Ledger.	**Branch Manager** stock viewing and Expense logging.	Supplier Invoice system and Purchase Order logic.
Sprint 3	**Corporate Quotation** pipeline and Payment tracking.	**CEO Approval** system for new users and transfers.	**Inter-branch Transfer** mechanism and Faulty stock logs.
Sprint 4	**Master Reports** and Branch-wise cash flow auditing.	Multi-branch monitoring and performance analytics.	Final Inventory stream testing and API optimization.

Table 2.2: Parallel development tasks by user role and sprint.

Chapter 3

Methodology

3.1 System Architecture

The ERP system is built on a three-tier architecture, designed to handle concurrent access from multiple geographical branches while maintaining data integrity.

- **Frontend:** A responsive web interface that provides role-specific dashboards for Executives, Branch Managers, Inventory Manager, Accounts and the CEO.
- **Backend:** Server-side logic (PHP) that manages session-based authentication, implements business rules for stock movements, and handles the multi-step Corporate Quotation pipeline.
- **Database:** A centralized MySQL relational database that serves as the ”single source of truth” for inventory, sales, and branch expenditures.

3.2 Database Design

The system's back-end is architected on a relational database normalized to **Third Normal Form (3NF)**, ensuring that every non-key attribute is functionally dependent solely on the primary key. This eliminates data redundancy and prevents update anomalies. The architecture utilizes complex relational mapping and **ACID-compliant transactions** to ensure that multi-branch operations are **atomic**; if any part of a stock movement or financial record fails, the entire transaction is rolled back to maintain absolute data consistency.

- **Identity and Access Control (RBAC):** A centralized **users** table serves as the core of the authentication system, linked via foreign keys to **roles** and **branches**. This structure enforces strict data isolation, where non-administrative users are restricted to records associated with their specific **branch id** through global query filtering.

- **Inventory Lifecycle and Traceability:** The schema implements a sophisticated tracking system through relational mapping between **purchase orders**, **product stock**, and **stock faulty items**. By utilizing a **Lot-based tracking** approach, the database links individual stock entries to their original **supplier invoice id**, allowing for precise FIFO (First-In-First-Out) cost auditing and profit margin calculation.

- **Decoupled Sales Architecture:** To handle diverse revenue streams, the database employs separate relational paths for **direct sales** and **corporate sales**. While retail transactions are optimized for speed in a flat structure, corporate sales utilize a tiered schema (quotations $\rightarrow$ orders $\rightarrow$ invoices $\rightarrow$ payments) to manage complex credit terms and partial payment tracking without data loss.

- **Logistics and Conflict Resolution:** The **product transfer** mechanism acts as a bridge between branch inventories. It utilizes a state-machine logic (Pending $\rightarrow$ Approved $\rightarrow$ Received) with **from branch** and **to branch** foreign keys. Triggers are implemented to ensure that stock is "locked" during the transfer process, preventing double-counting or negative inventory levels.

- **Financial Integrity and Audit Trails:** Every transaction that impacts the **cash in hand** or **customer ledgers** is recorded in an immutable **transaction logs** table. By enforcing ON DELETE RESTRICT constraints on primary entities, the database ensures that no historical financial data can be purged if dependent sales or expense records exist, providing a permanent audit trail for the Accounts department.

ER Diagram

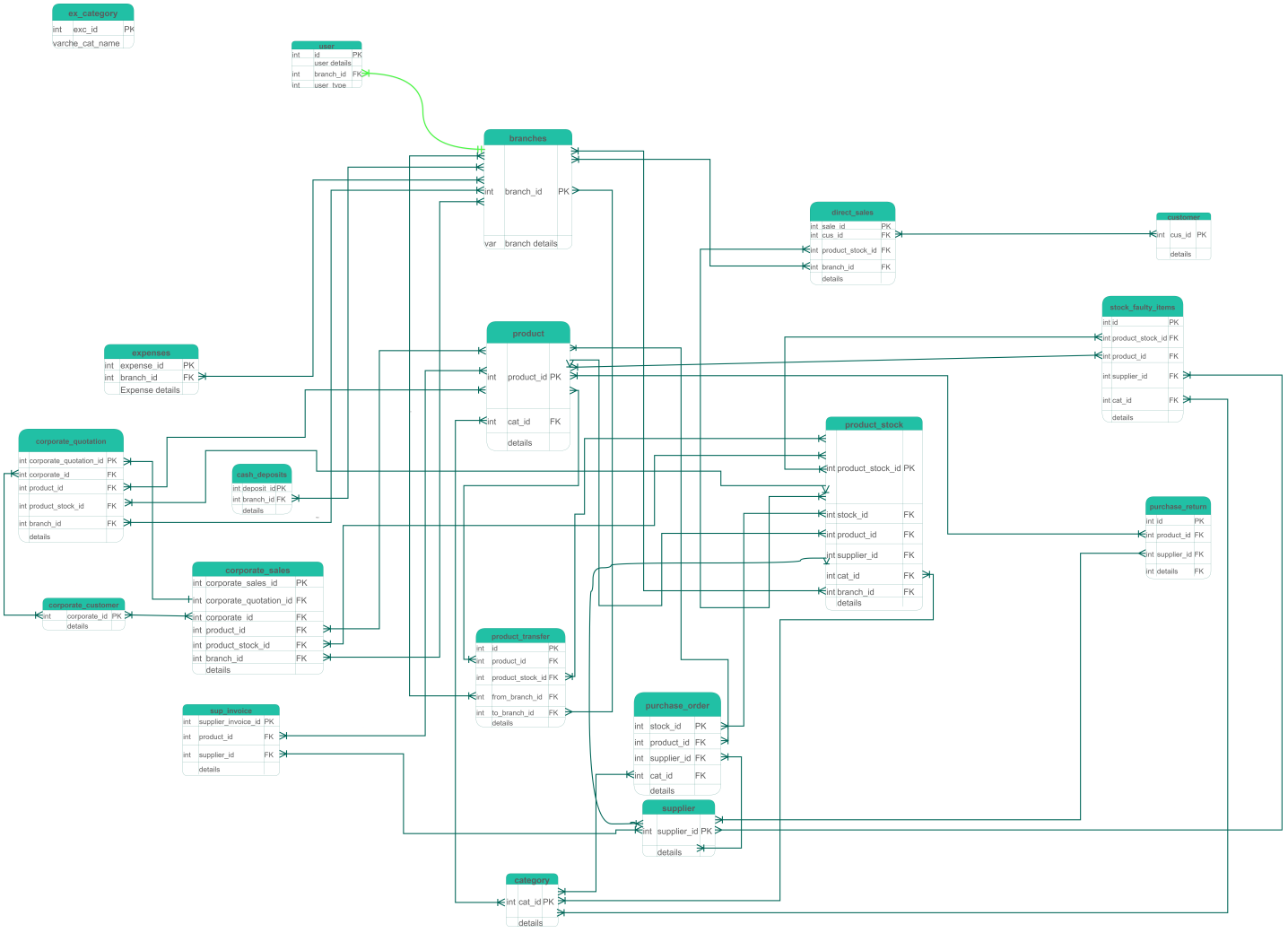


Figure 3.1: Detailed ER Diagram showing the relationships between Inventory, Sales, and Branch entities.

3.3 Operational Workflow

The operational logic is strictly governed by user roles to ensure security and financial accountability:

1. **Procurement and Inwarding:** The **Inventory Manager** adds products from suppliers. The system automatically generates a **sup_invoice**, calculates the initial **due_amount**, and assigns a unique **lot_no** for tracking.
2. **Stock Distribution:** The **Inventory Manager** initiates transfers between branches (e.g., HQ to Sylhet). This requires status updates within the **product_transfer** table to ensure stock is never ”in-transit” indefinitely.
3. **Sales Execution:**
   - **Executives** process retail sales, which immediately updates the **product_stock** and customer ledger.
   - **Corporate Sales** follow a Quotation $\rightarrow$ Approval $\rightarrow$ Billing cycle.
4. **Financial Audit:** The **Accounts** role monitors supplier dues and branch-level **cash_deposits**, while the **Branch Manager** logs operational **expenses**.
5. **Administrative Oversight:** The **CEO** manages the user lifecycle, approving or declining registration requests with automated mail notifications, and reviews consolidated performance reports.

Use Case diagram

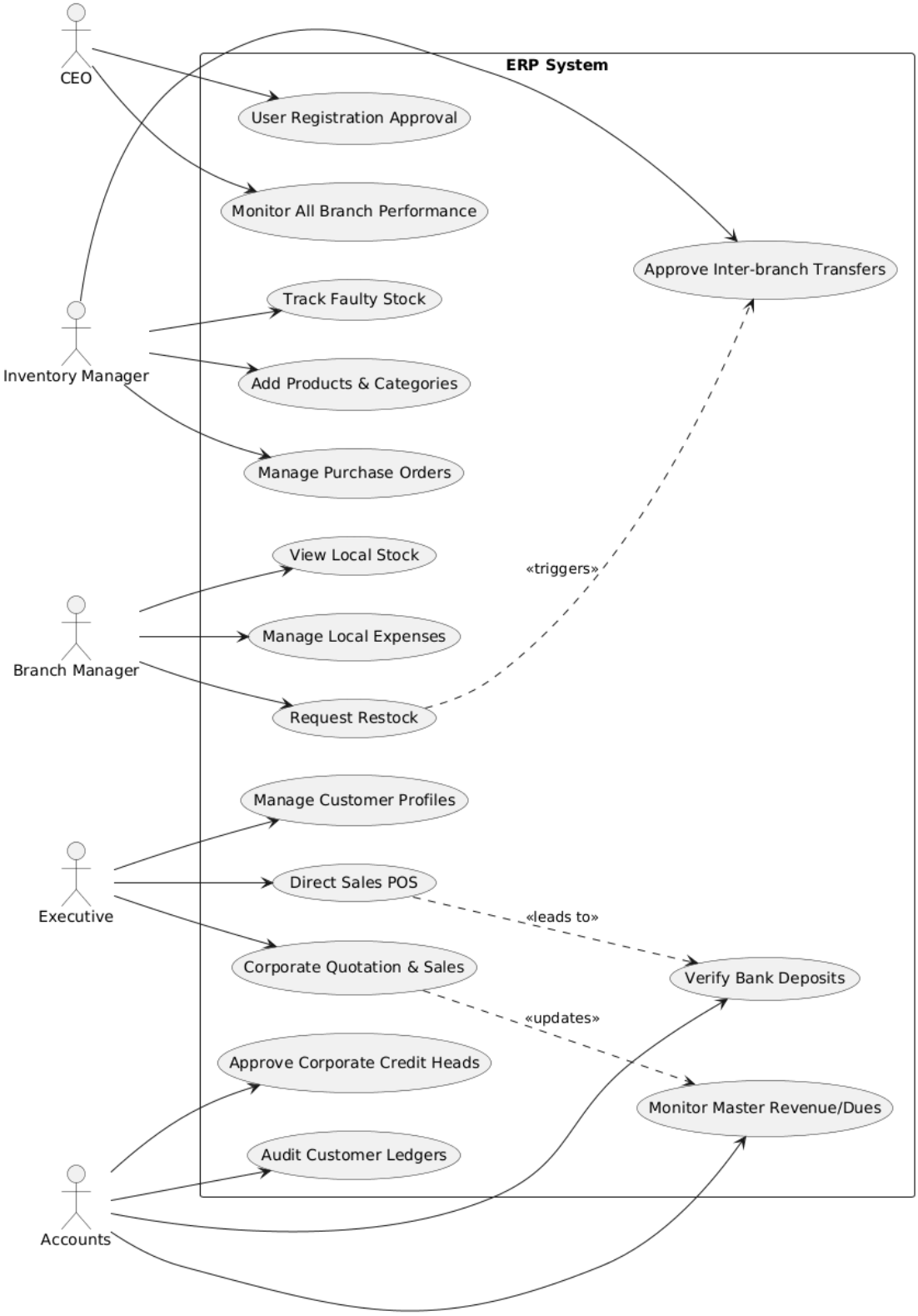


Figure 3.2: Use Case Diagrams

Chapter 4

Results and Discussion

4.1 Functional Achievement and Evaluation

The implementation of the ERP system has successfully addressed the challenges of fragmented data across multiple branches. The system's performance was evaluated based on its ability to maintain data integrity and enforce role-based security.

4.1.1 Operational Efficiency

The automation of several manual processes has yielded significant improvements in operational speed:

- **Financial Documentation:** Invoices for supplier purchases are now auto-generated with calculated due and paid amounts, reducing accounting errors.
- **Real-time Stock Visibility:** Branch Managers can now view branch-specific stock levels instantaneously, a task that previously required manual auditing.
- **Stock Movement:** The inter-branch transfer module ensures that stock levels are adjusted in both the source and destination branches simultaneously, preventing "ghost inventory".

4.1.2 Security and Access Control

The Role-Based Access Control (RBAC) was tested against five user roles (CEO, Inventory Manager, Branch Manager, Accounts, and Executive). Key security outcomes

include:

- **Closed Registration Loop:** No user can access the system without explicit CEO approval.
- **Unauthorized Access Prevention:** Testing confirmed that users attempting to bypass their dashboard via URL manipulation were redirected to an ”Unauthorized” error page.
- **Notification System:** The integration of SMTP mail services ensures that users are immediately notified of their account status, improving the onboarding experience.

4.2 Project Screenshots and Explanation

The following figures illustrate the primary interfaces of the ERP system and explain the underlying business logic and technical implementation for each module.

User Registration and Submission:

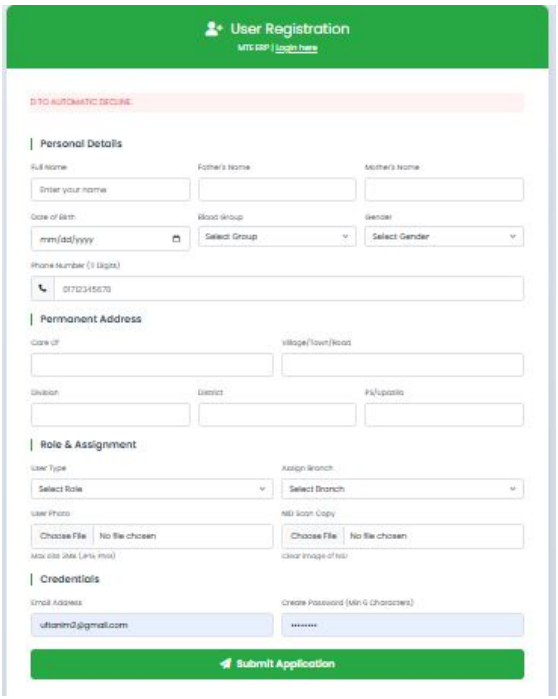


Figure 4.1: User Registration Interface and Information Submission Confirmation

4.2.1 CEO Interface

The following figures illustrate the CEO's administrative tools, providing high-level oversight of financial health, user management, and inter-branch logistics.

Administrative Approval

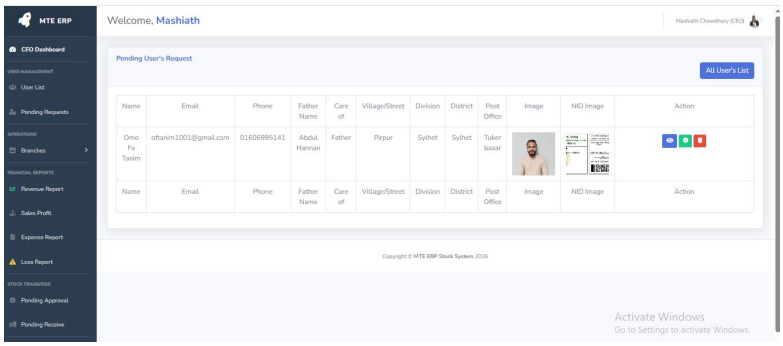


Figure 4.2: CEO Approval Interface

CEO Dashboard and Monitoring:

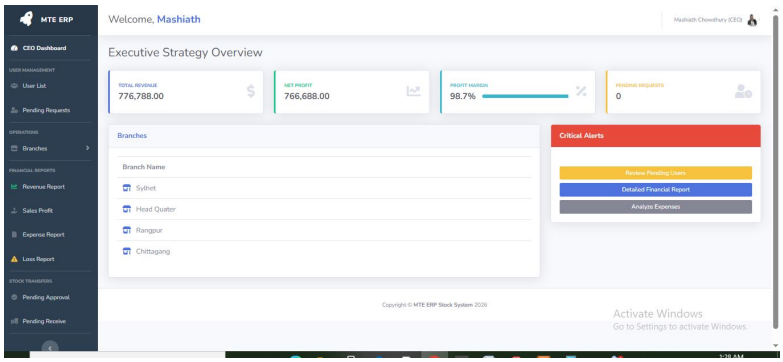


Figure 4.3: CEO Panel: Financial Summaries, Branch Status, and Alert Management

User Management and Access Control:

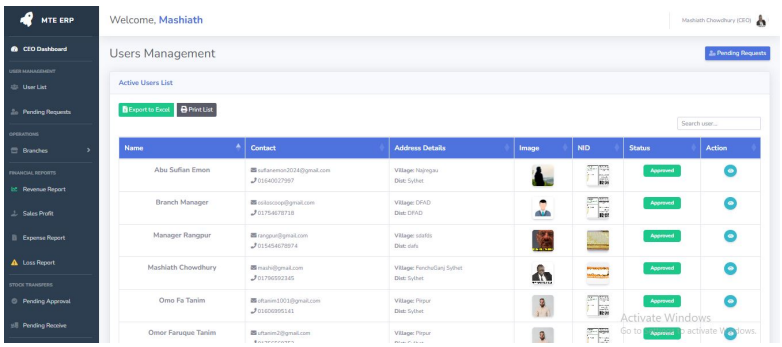


Figure 4.4: Active User List with Role and Branch Designations

Branch Oversight and Management:

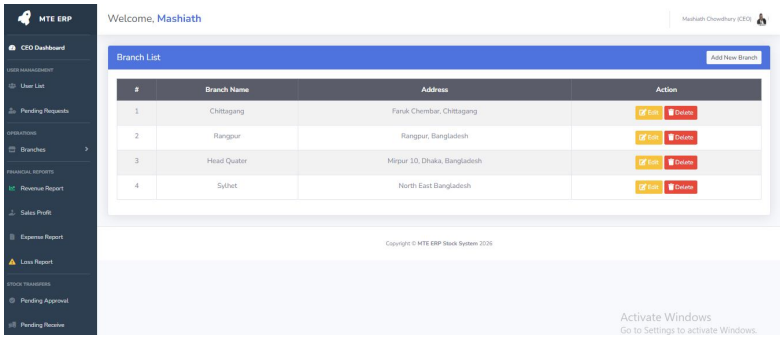


Figure 4.5: Centralized Branch List with Management Controls

Consolidated Revenue Reporting:

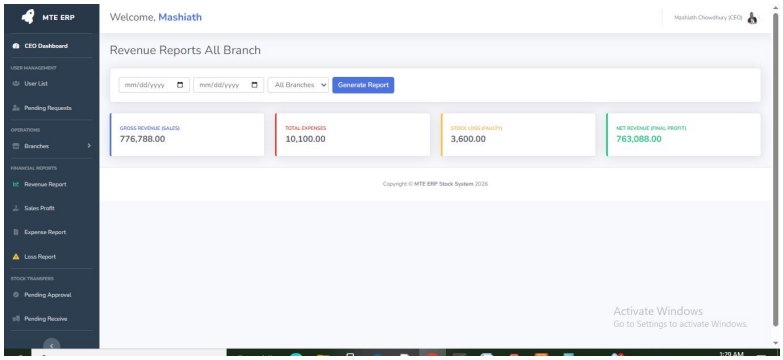


Figure 4.6: Multi-branch Revenue Report

Sales Performance Report:

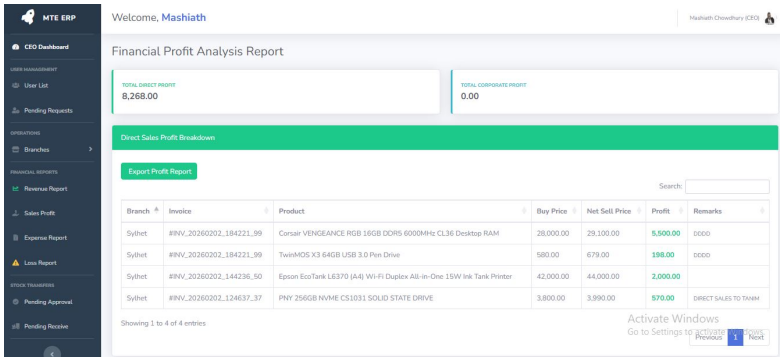


Figure 4.7: Detailed Sales Report and Transaction Logs

Expense Monitoring and Audit:

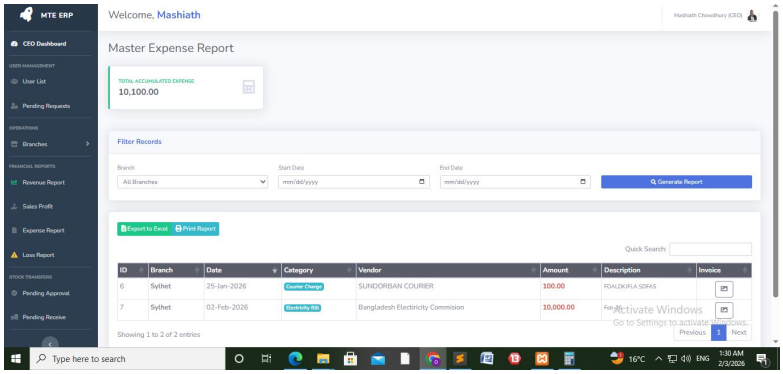


Figure 4.8: Consolidated Expense Report for Organizational Cost Analysis

Loss Analysis:

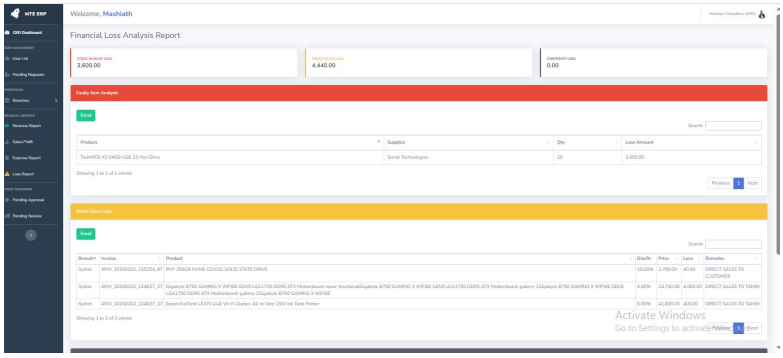


Figure 4.9: Profit and Loss Report for Organizational Performance Evaluation

Pending Transfer Approval:

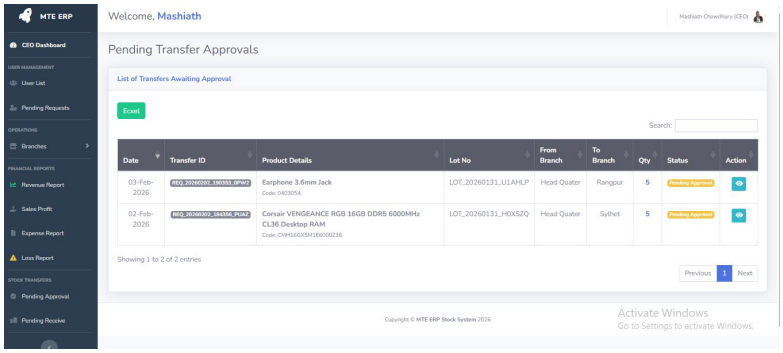


Figure 4.10: CEO Interface for Reviewing and Authorizing Inter-Branch Stock Transfers

Pending Transfer Receipt:

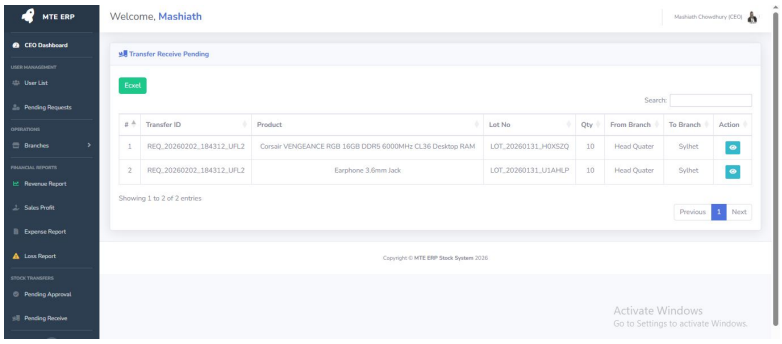


Figure 4.11: Monitoring View for Stock Transfers Awaiting Final Receipt Confirmation

4.2.2 Branch Manager Interface

This section presents the operational interfaces utilized by the Branch Manager for inventory, sales, and financial oversight at the branch level.

Local Branch Stock:

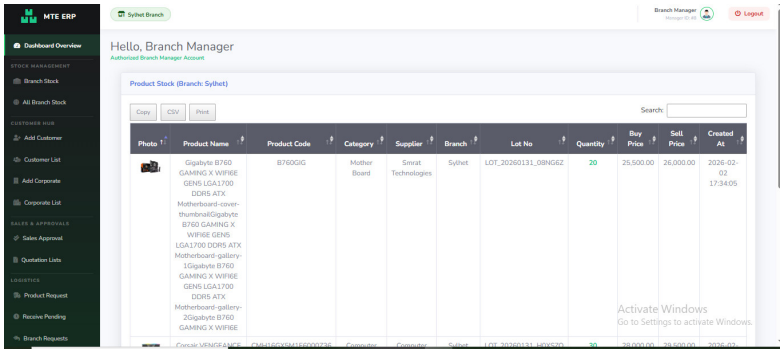


Figure 4.12: Current Inventory Levels for the Specific Branch

Global Inventory Overview:

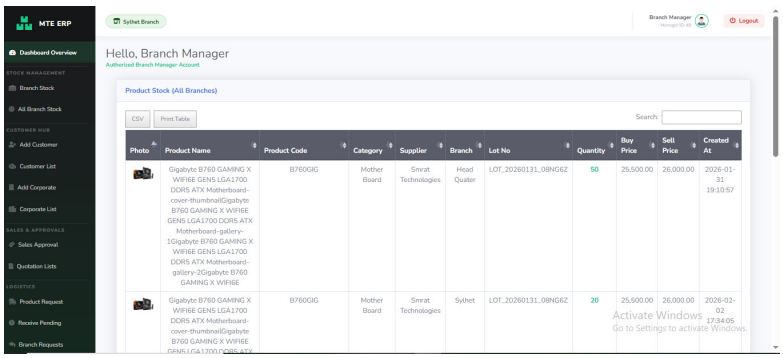


Figure 4.13: Consolidated View of Stock Levels Across All Branches

Customer Registration:

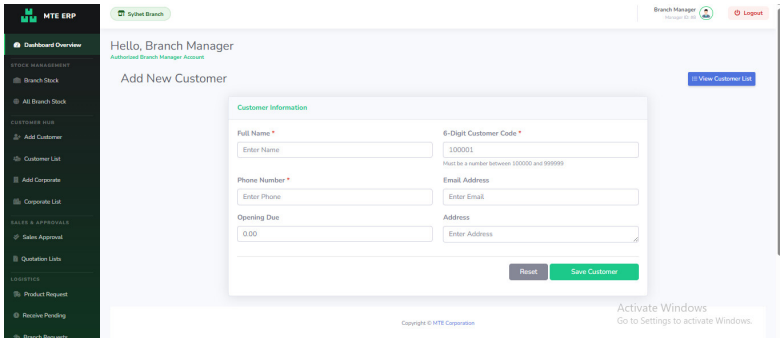


Figure 4.14: Interface for Adding New Customer Records

Customer Directory:

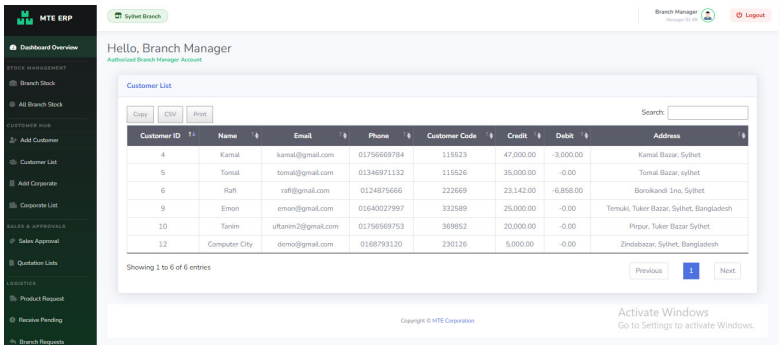


Figure 4.15: List of Registered Customers and Contact Information

Direct Sales Approval:

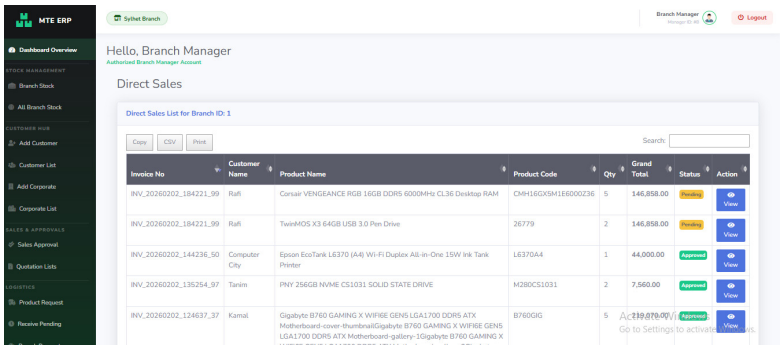
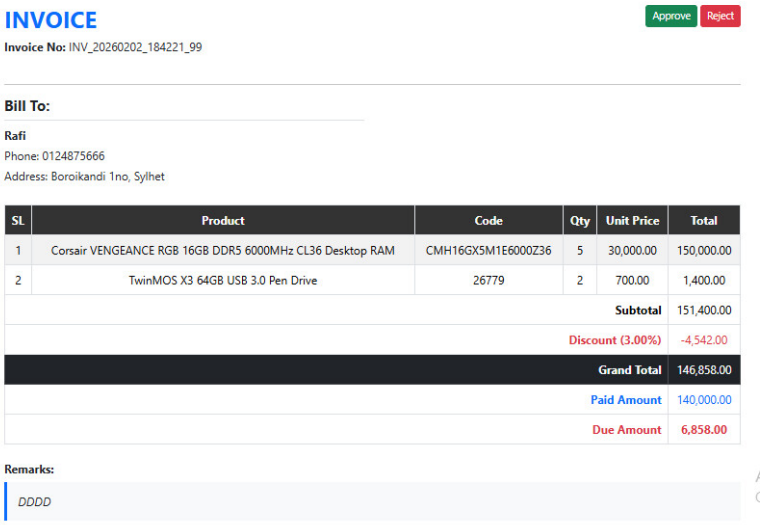


Figure 4.16: Management Interface for Authorizing Retail Sales

Sales Invoice:



INVOICE

Approve Reject

Invoice No: INV_20260202_184221_99

Bill To:

Rafi
Phone: 0124875666
Address: Boroikandi 1no, Sylhet

SL	Product	Code	Qty	Unit Price	Total
1	Corsair VENGEANCE RGB 16GB DDR5 6000MHz CL36 Desktop RAM	CMH16GX5M1E6000Z36	5	30,000.00	150,000.00
2	TwinMOS X3 64GB USB 3.0 Pen Drive	26779	2	700.00	1,400.00
				Subtotal	151,400.00
				Discount (3.00%)	-4,542.00
				Grand Total	146,858.00
				Paid Amount	140,000.00
				Due Amount	6,858.00

Remarks:

DDDD

Figure 4.17: Generated Invoice Template for Completed Sales

Corporate Quotation:

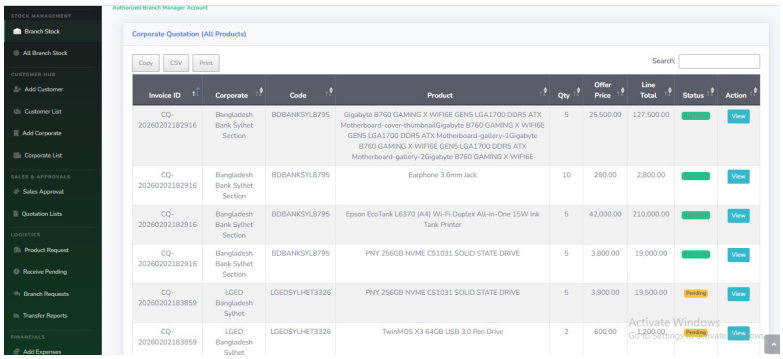


Figure 4.18: Interface for Creating Corporate Sales Quotations

Product Request:

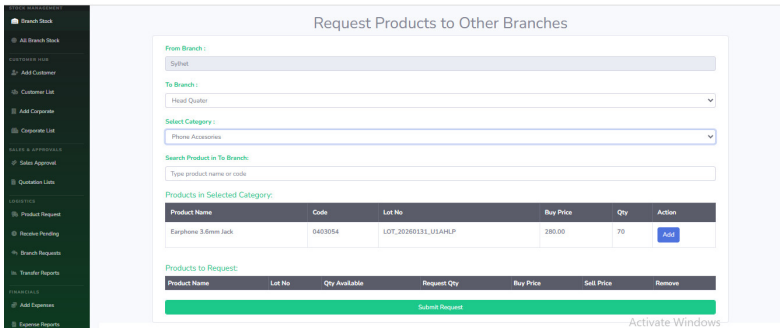


Figure 4.19: Internal Request System for Procurement of New Items

Transfer Receipt Management:

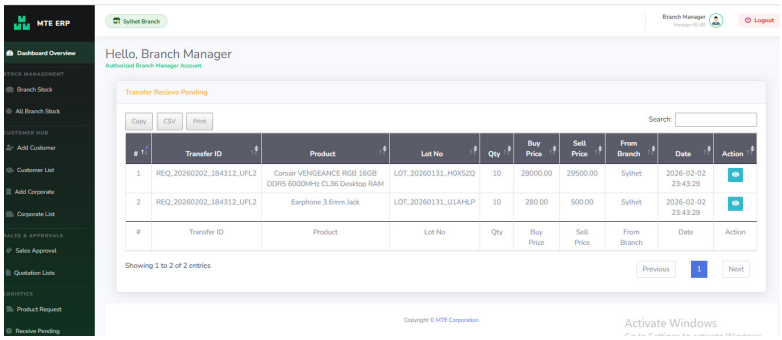


Figure 4.20: Tracking Interface for Incoming Inter-Branch Transfers

Branch Request Logs:

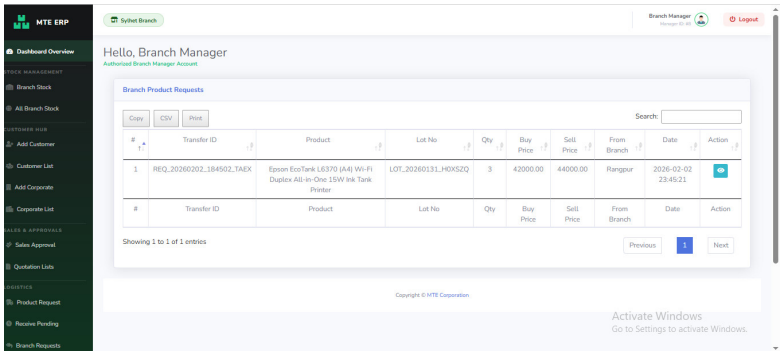


Figure 4.21: Overview of Pending and Fulfilled Product Requests

Transfer Reports:

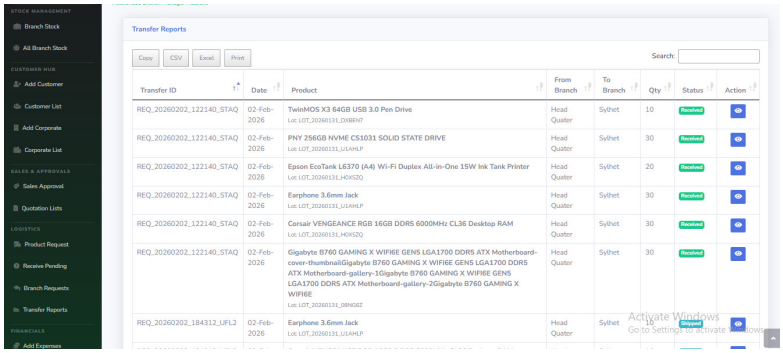


Figure 4.22: Detailed History of Stock Movements Between Branches

New Expense Entry:

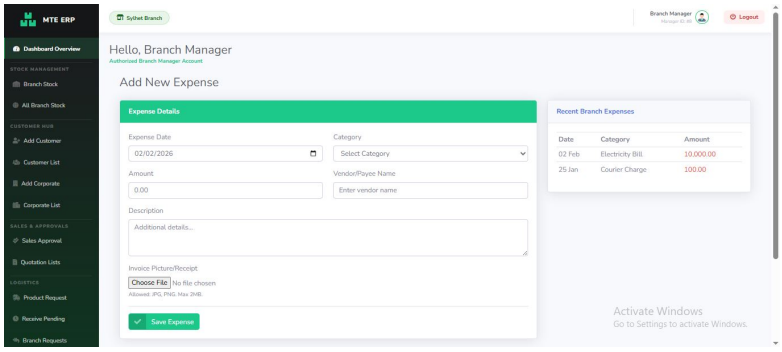


Figure 4.23: Submission Form for Local Branch Expenditures

Expense Management:

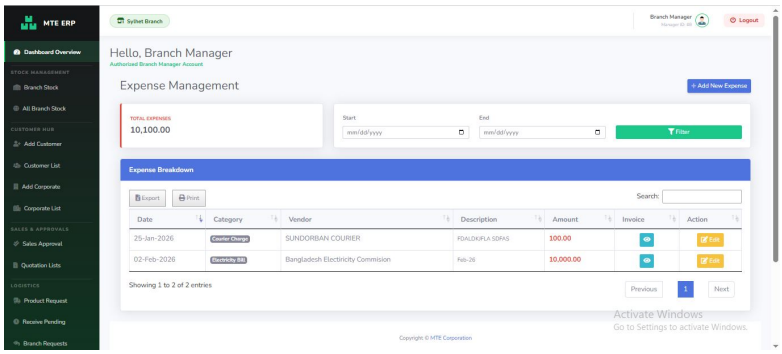


Figure 4.24: Tracking and Categorization of Branch-Level Expenses

4.2.3 Executive User Interface

This section displays the operational tools used by the Executive for point-of-sale activities, corporate client management, and real-time stock tracking.

Direct Sales POS:

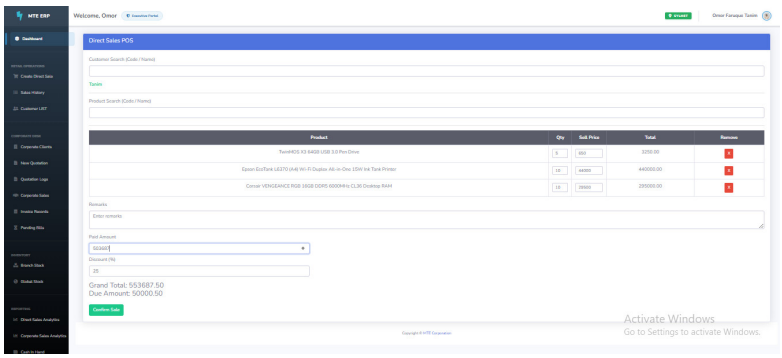


Figure 4.25: Point-of-Sale interface for retail transactions

Direct Sales List:

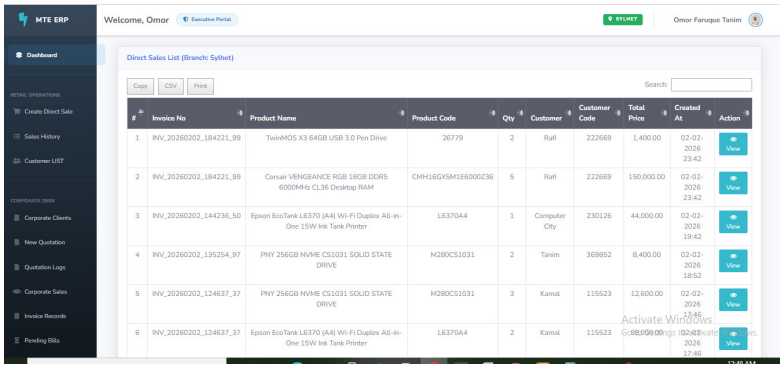


Figure 4.26: Branch-specific retail transaction history

Sales Invoice:

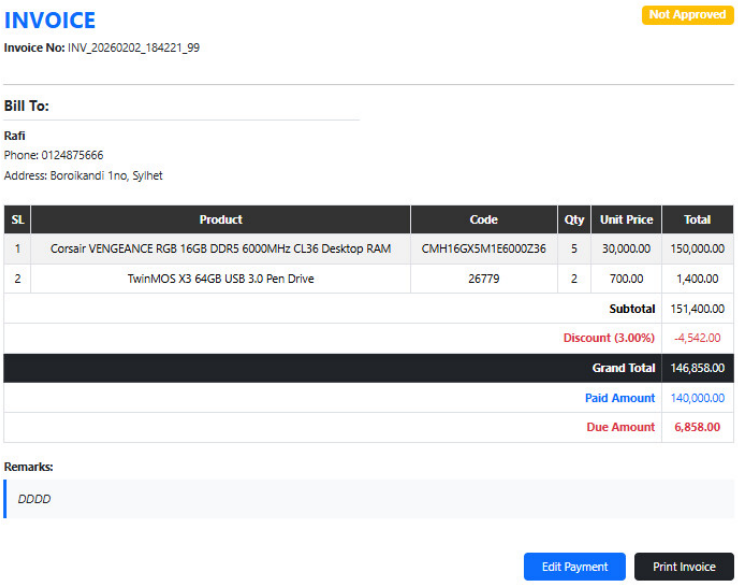


Figure 4.27: Standard invoice generation for customers

Update Invoice Payment:

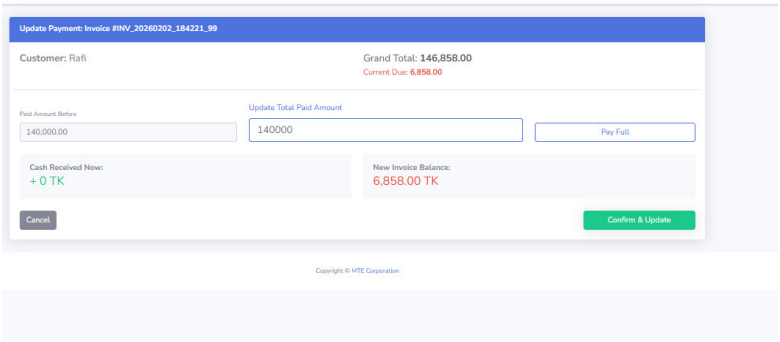


Figure 4.28: Interface for updating transaction payment status

Customer Directory:

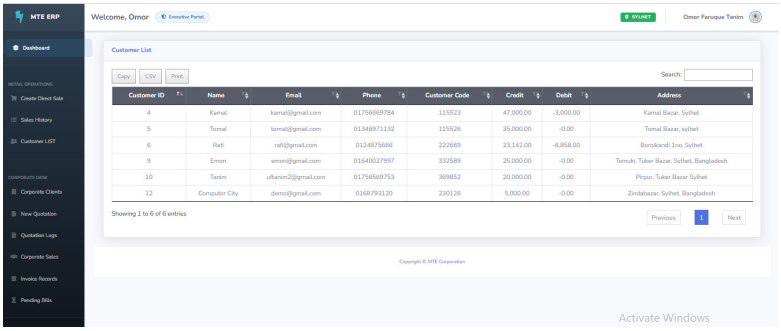


Figure 4.29: Management list for registered retail customers

Approved Corporate Clients:

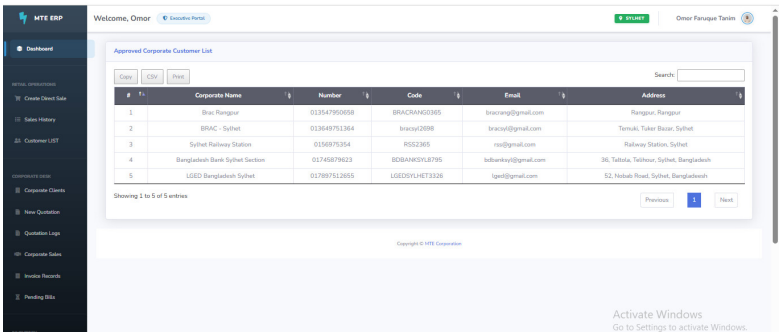


Figure 4.30: Verified list of corporate account entities

Create Corporate Quotation:

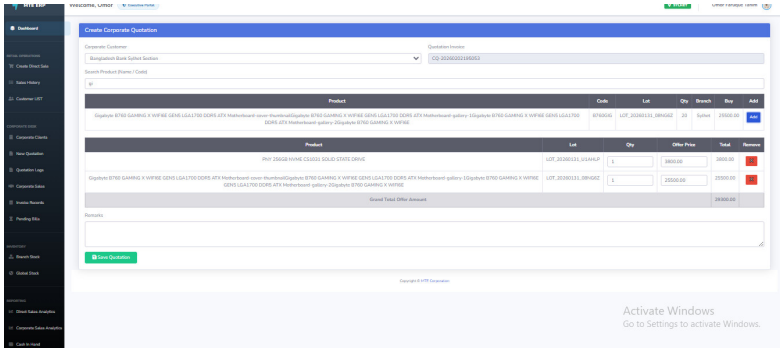


Figure 4.31: Generation form for corporate price quotations

Corporate Quotation List:

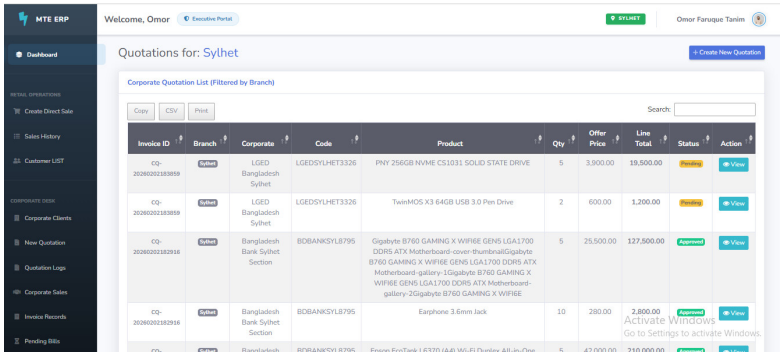


Figure 4.32: Branch record of all issued corporate quotes

Corporate Sales:

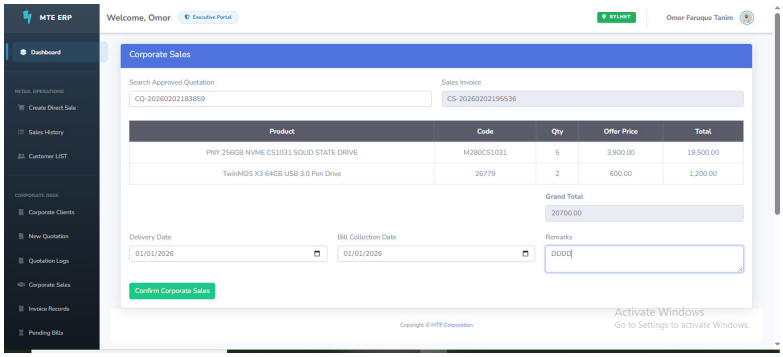


Figure 4.33: Interface for processing corporate-tier sales

Corporate Billing Records:

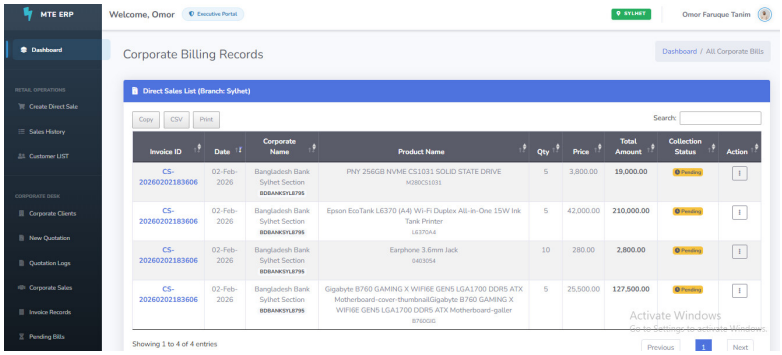


Figure 4.34: Log of billing cycles for corporate clients

Pending Corporate Bills:

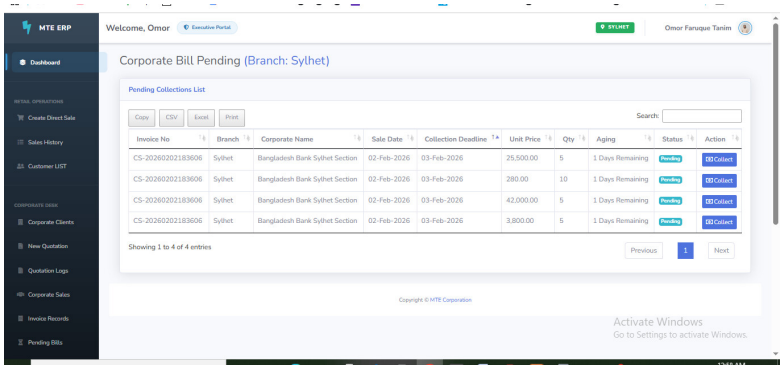


Figure 4.35: Tracking of outstanding corporate invoices

Branch Product Stock:

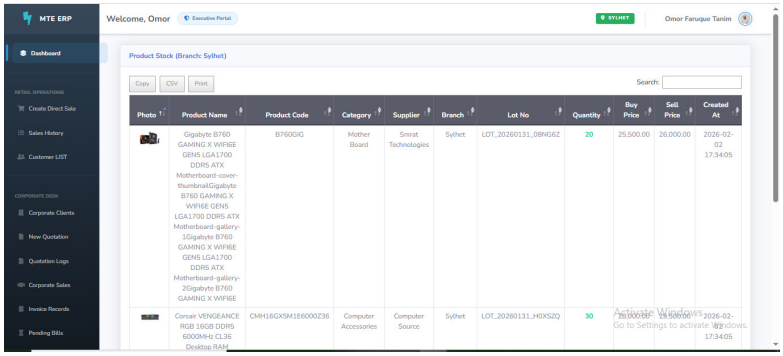


Figure 4.36: Current inventory availability for the local branch

Global Product Stock:

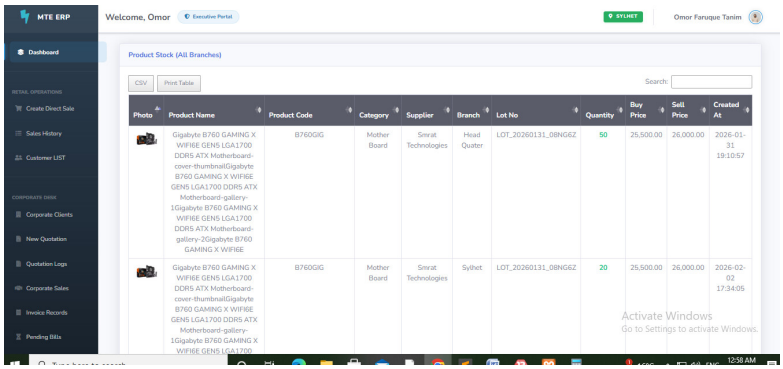


Figure 4.37: Stock levels across all organizational branches

Direct Sales Report:

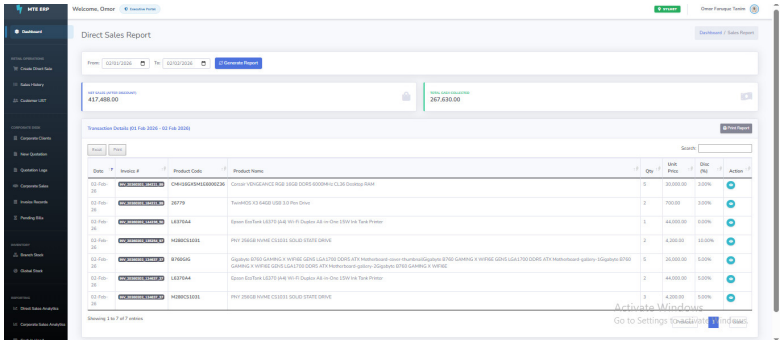


Figure 4.38: Detailed analytics of retail sales performance

Corporate Strategic Report:

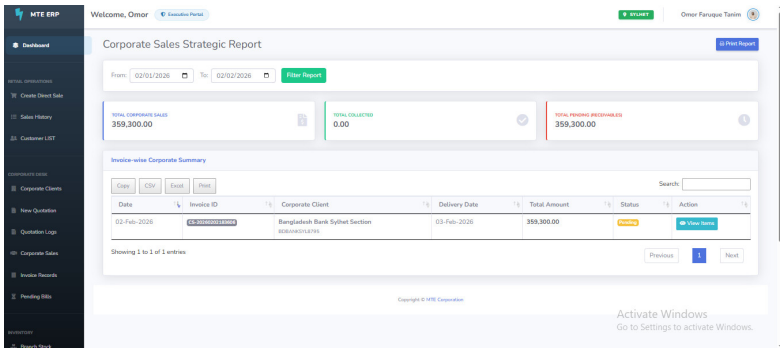


Figure 4.39: Performance report for corporate sales accounts

Cash Report:

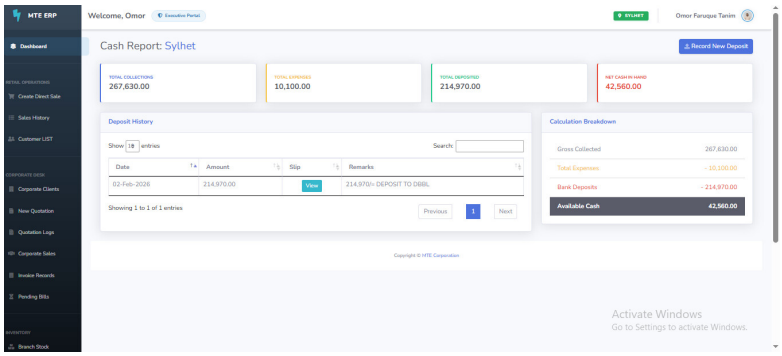


Figure 4.40: Summary of cash flow and daily collections

4.2.4 Inventory Manager Interface

This section covers the tools used for supply chain management, including stock tracking, procurement, supplier relations, and inter-branch transfers.

Inventory Dashboard:

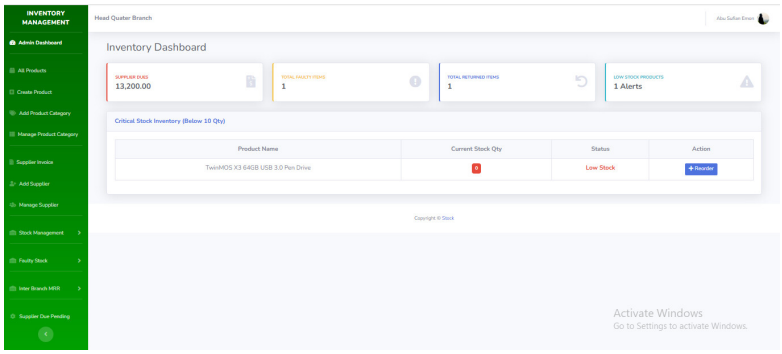


Figure 4.41: Dashboard featuring critical stock alerts and inventory status

Product Directory:

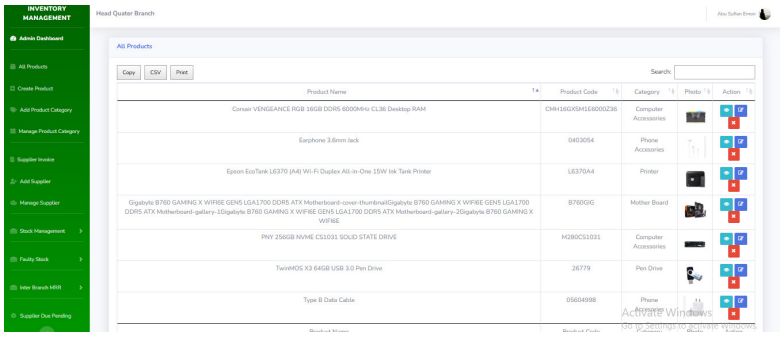


Figure 4.42: Master list of all products registered in the system

Add New Product:

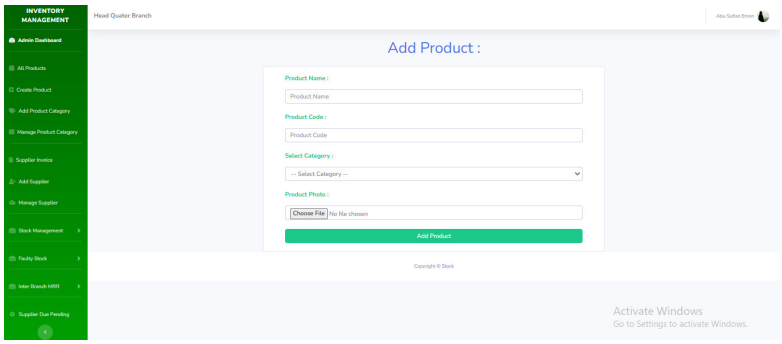


Figure 4.43: Form for adding new items to the inventory database

Product Category Management:

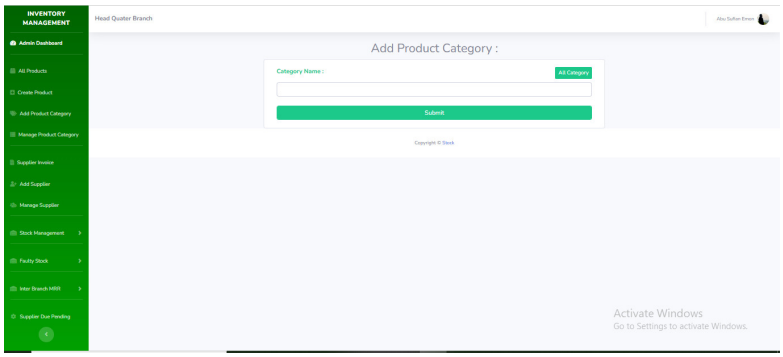


Figure 4.44: Interface for creating and organizing product categories

Category List:

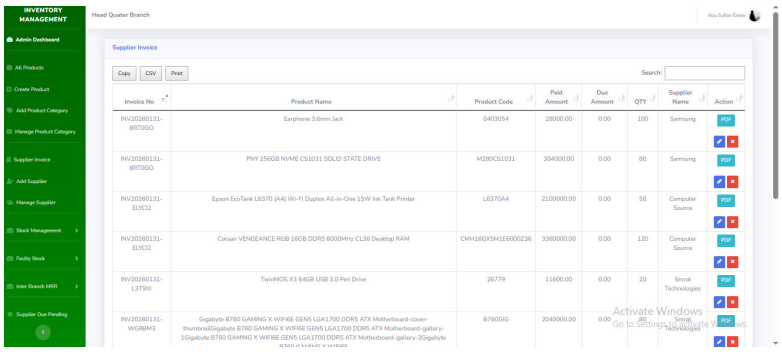


Figure 4.45: Overview of all active product classification categories

Supplier Invoice List:

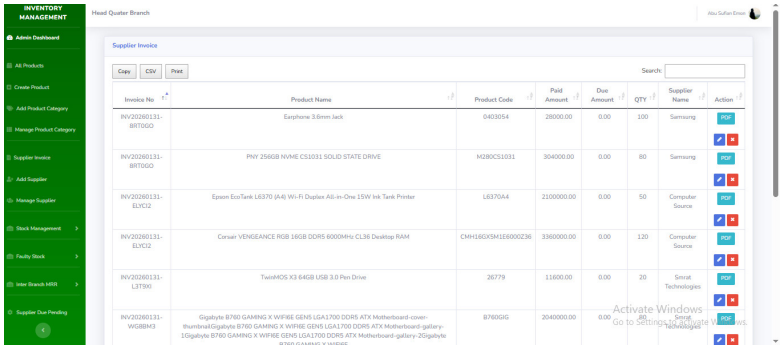


Figure 4.46: Record of all invoices received from external suppliers

Supplier Invoice View:

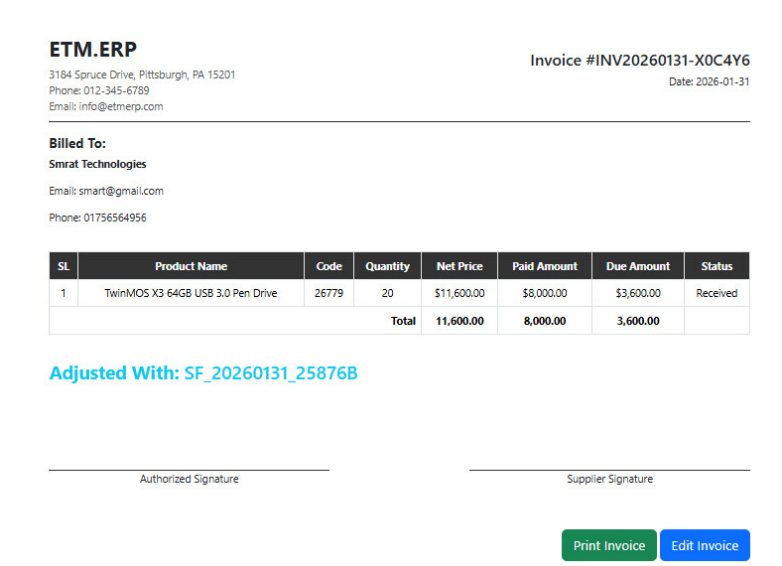


Figure 4.47: Detailed view of specific supplier billing documents

Payment Status Update:

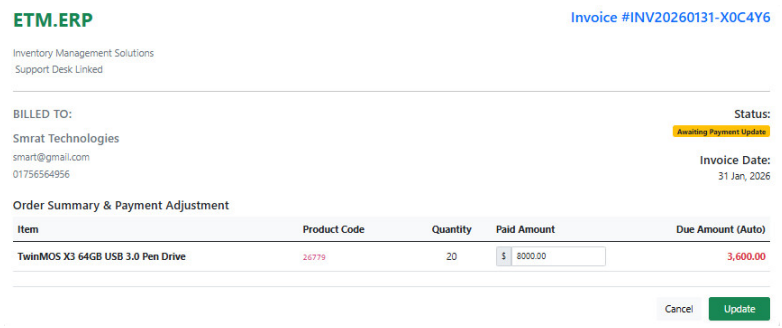


Figure 4.48: Tool for updating paid amounts on supplier invoices

Supplier Registration:

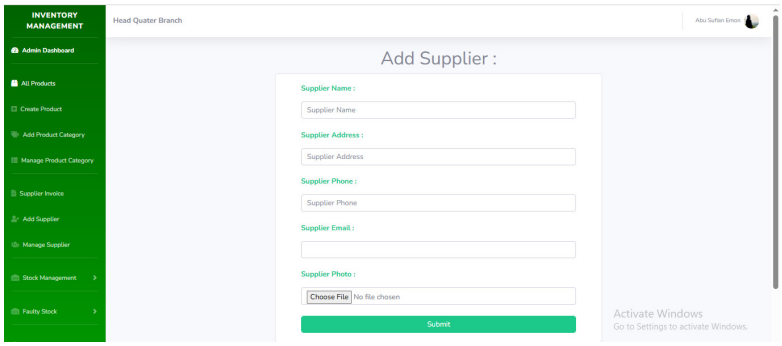


Figure 4.49: Interface for adding new supplier contact and billing data

Supplier Master List:

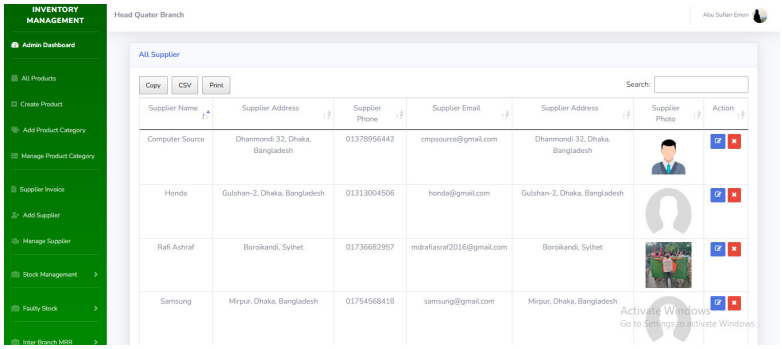


Figure 4.50: Database of all approved organizational suppliers

Create Purchase Order:

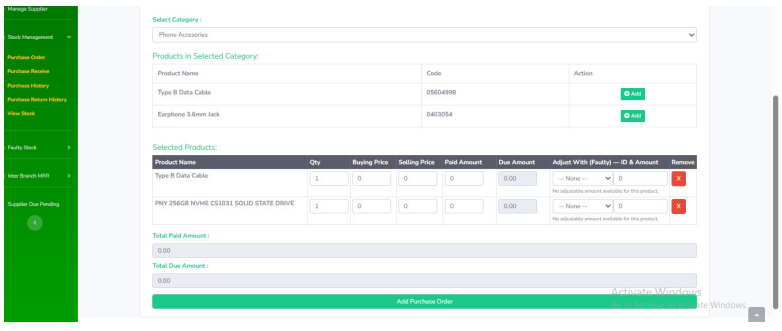


Figure 4.51: System for generating new stock purchase orders

Pending Purchase Receipts:

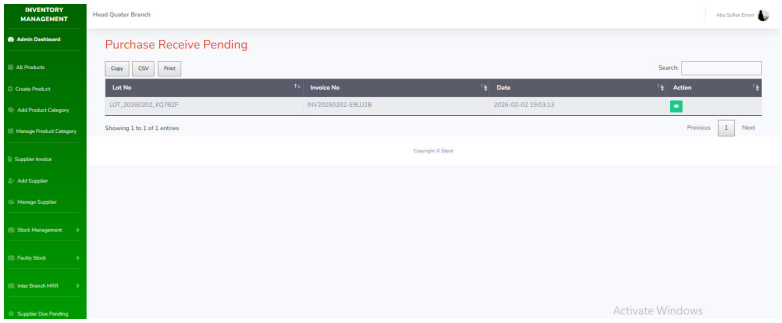


Figure 4.52: Tracking for orders awaiting warehouse arrival

Stock Receipt and Returns:

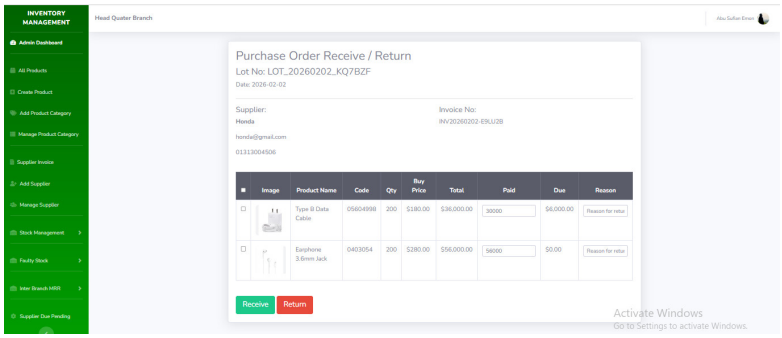


Figure 4.53: Interface for receiving stock or processing returns to suppliers

Purchase Return History:

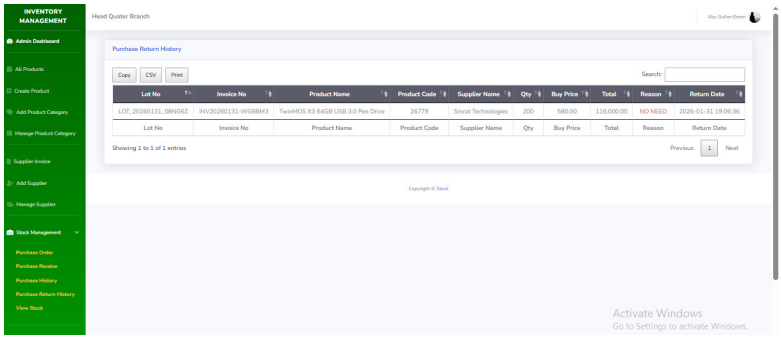
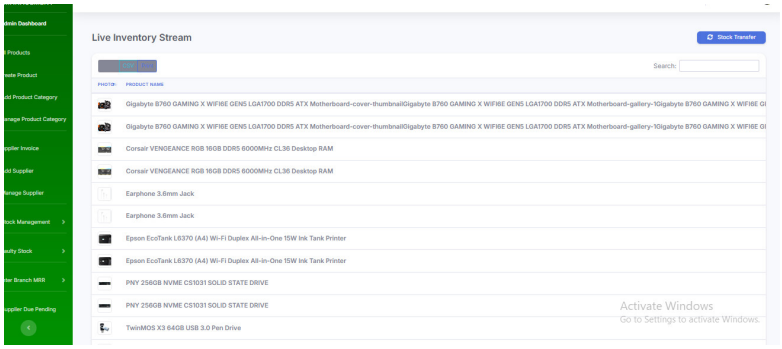


Figure 4.54: Logged record of all items returned to suppliers

Live Inventory Stream:


Live Inventory Stream
Stock Transfer
Search:
Gigabyte B760 GAMING X WIFI6E GEN5 LGA1700 DDR5 ATX Motherboard-cover-thumbnailGigabyte B760 GAMING X WIFI6E GEN5 LGA1700 DDR5 ATX Motherboard-gallery-1

New Product Transfer:

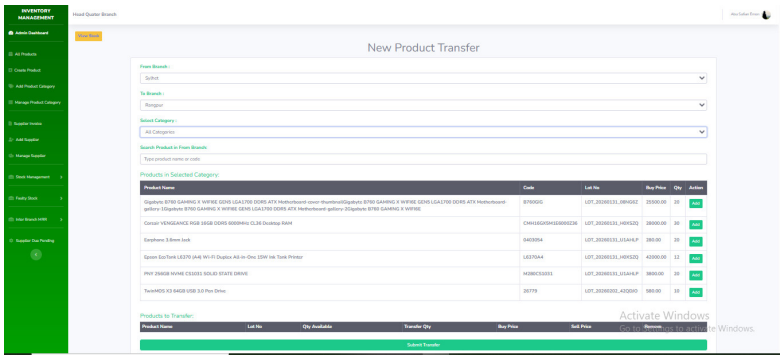


Figure 4.58: Interface for initiating stock transfers between branches

Branch Transfer Requests:

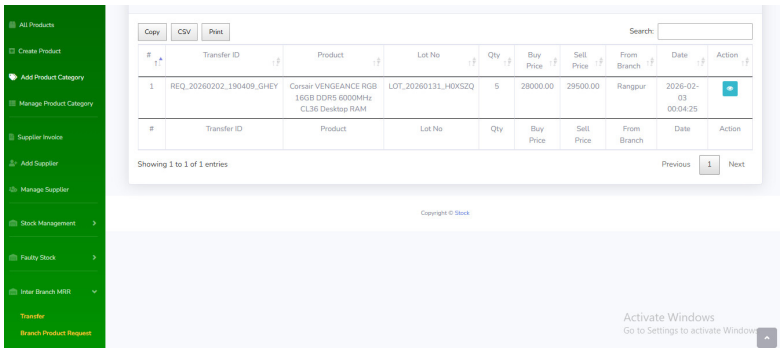


Figure 4.59: Management of incoming stock requests from various branches

4.2.5 Accounts Interface

This section outlines the financial oversight tools used by the Accounts department for auditing, ledger management, and organization-wide cash flow tracking.

Accounts Dashboard:

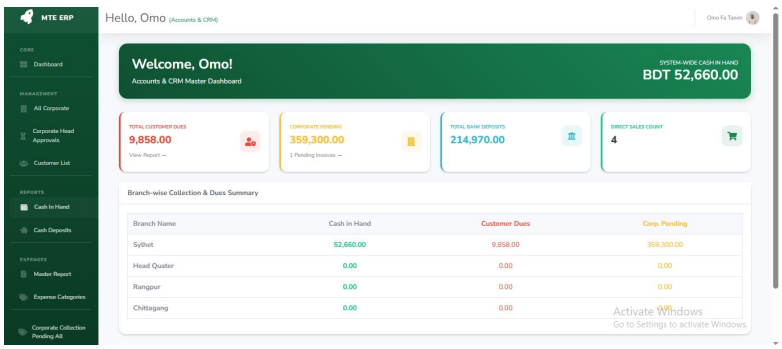


Figure 4.60: Financial overview featuring key liquidity metrics and alerts

Corporate Approval Pending:

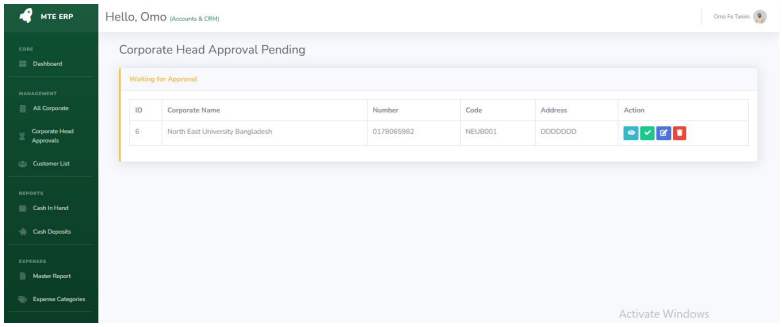


Figure 4.61: Review queue for authorizing new corporate account heads

Requested Customer Directory:

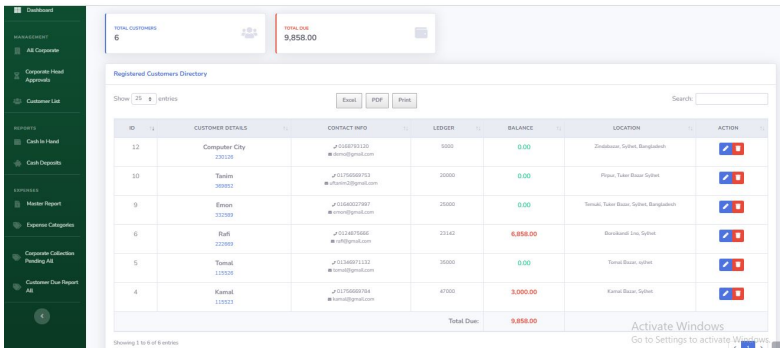


Figure 4.62: List of customers awaiting account verification or credit approval

Customer Ledger Management:

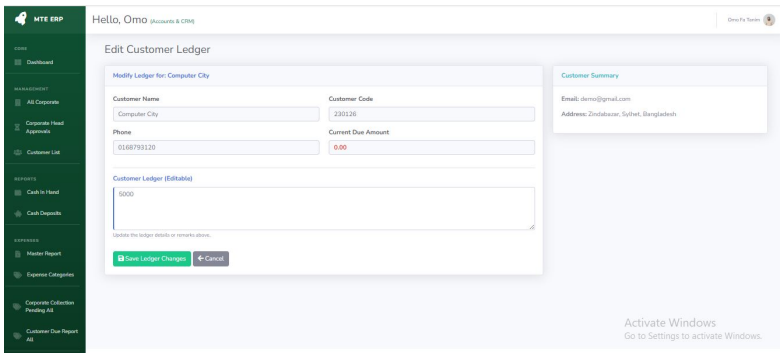


Figure 4.63: Interface for auditing and editing customer financial statements

Branch Cash-in-Hand:



Figure 4.64: Real-time monitoring of physical cash available at each branch

Branch Cash Deposits:

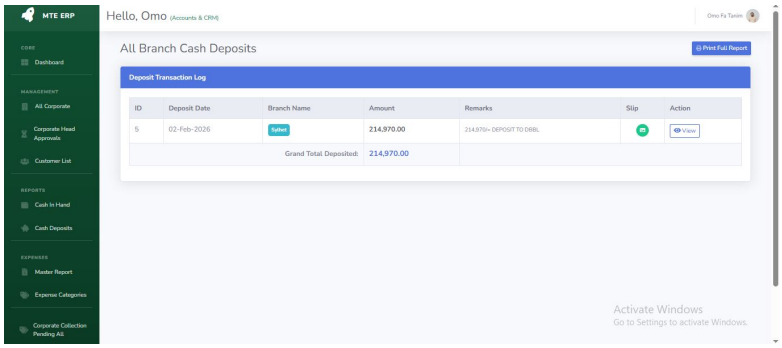


Figure 4.65: Tracking of bank deposits and cash transfers from branches

Master Expense Report:

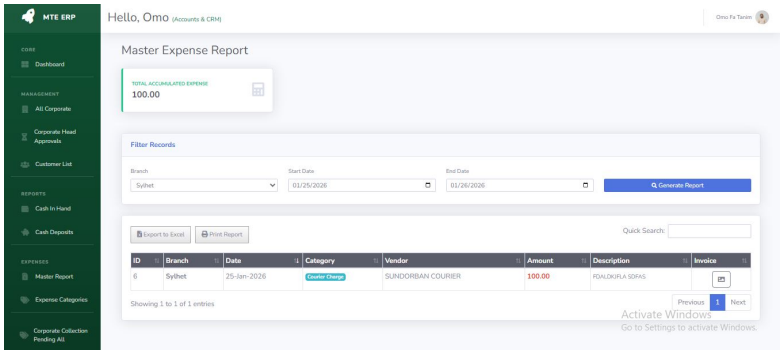


Figure 4.66: Consolidated audit of all organizational expenditures

Master Corporate Collections:

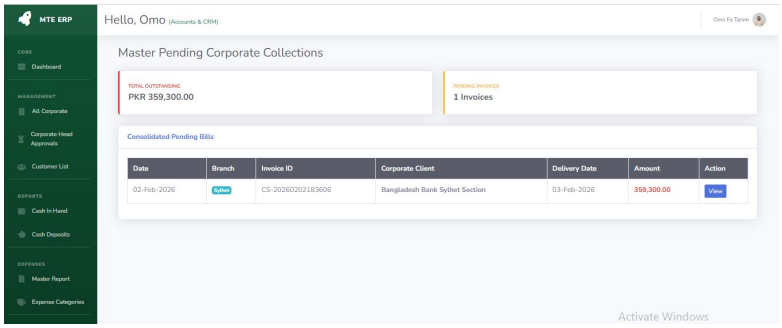


Figure 4.67: Tracking of pending payments from all corporate clients

Master Sales and Due Report:

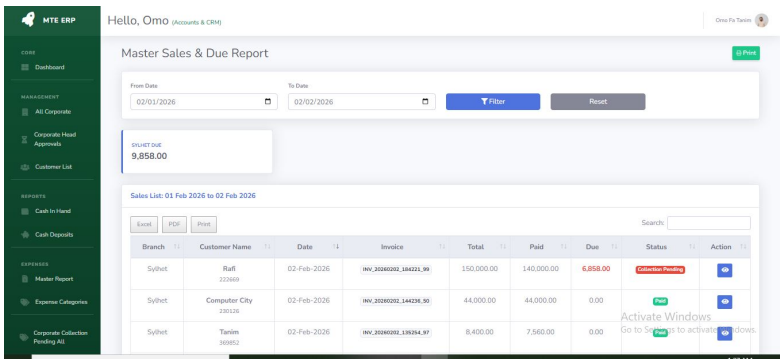


Figure 4.68: Global report of total sales versus outstanding receivables

Chapter 5

Conclusion

5.1 Project Summary

The developed ERP system successfully integrates core business operations including sales, inventory, and branch management into a unified, centralized platform. By implementing a robust Role-Based Access Control (RBAC) mechanism, the system ensures that the **CEO, Inventory Manager, Branch Manager, Accounts, and Executive** roles operate within their designated functional boundaries. This architecture not only enhances data security but also significantly improves operational efficiency by providing real-time visibility across geographically distributed branches like Sylhet and Rangpur.

The transition from manual or fragmented processes to this automated solution has demonstrated a measurable reduction in human error. Automation of repetitive tasks such as invoice generation, payment tracking via the `sup_invoice` table, and inter-branch stock reconciliation has streamlined the internal supply chain. Furthermore, the centralized relational database serves as a ”single source of truth,” enabling seamless communication between departments and providing stakeholders with accurate data for informed decision-making.

5.2 Project Contributions and Practical Impact

As a completed final-year project, this system provides several key contributions to the organization’s digital infrastructure:

- **Centralized Governance:** The CEO now possesses a dedicated dashboard to audit user registrations and monitor global revenue, ensuring high-level account-

ability.

- **Optimized Supply Chain:** The Inventory Manager can track product lifecycles from procurement to branch-level distribution, significantly reducing the risks of stock-outs or ghost inventory.
- **Financial Traceability:** The Accounts and Managerial modules provide a transparent record of supplier dues and branch expenses, facilitating more accurate profit-and-loss reporting.
- **Scalable Architecture:** The modular design of the backend and the normalized database schema allow the organization to add new branches or product categories with minimal technical overhead.

5.3 Final Closing Statement

In conclusion, this project demonstrates the effective application of software engineering principles ranging from database normalization and Object-Oriented Programming (OOP) to Agile Scrum methodologies. The resulting ERP system is not merely a tool for recording transactions but a strategic asset that enhances transparency, accountability, and scalability. This project successfully achieves all its primary objectives, providing a robust, lightweight, and cost-effective roadmap for the organization's long-term operational success.